



UNIVERSITETI I EVROPËS JUGLINDORE
УНИВЕРЗИТЕТ НА ЈУГОИСТОЧНА ЕВРОПА
SOUTH EAST EUROPEAN UNIVERSITY

Application for approval of Master Thesis

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To: Teaching Scientific Council
Faculty of Contemporary Sciences & Technologies
South East European University

Request

Dear Members of the Teaching Scientific Council,

My name is Diellza Qerimi, a master student of Data Science program of the Faculty of Contemporary Sciences and Technologies, with the student ID 127828.

I am writing to formally request your approval of my master's thesis proposal, titled **"Personalized Skincare Recommendations Using RAG and Large Language Models"**. This thesis will build a skincare recommendation system that leverages user reviews and product information, combining real consumer insights with the power of Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs). With the rapid growth of both the beauty industry and artificial intelligence, this research sits at the ideal intersection of the two, addressing a real-world challenge while contributing academically by demonstrating how emerging AI technologies can be tailored to domain-specific applications.

Mentor:
Asst. Prof. Dr. Nuhi Besimi

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Diellza Qerimi

Working title of the thesis:

“Personalized Skincare Recommendations Using RAG and Large Language Models”

Translation of the working title in Albanian:

“Rekomandime të personalizuar të produkteve të kujdesit të lëkurës duke përdorur RAG dhe Large Language Models”

Translation of the working title in Macedonian:

“Персонализирани препораки за производи за нега на кожата со користење на RAG и Large Language Models”

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1. Research field

Over the past few years, AI has shifted from a niche topic to a mainstream priority for individuals, companies, and entire sectors. Nearly every industry is experimenting with it—sometimes in surprising ways, like fashion brands using AI-generated models in their campaigns. Adoption isn't uniform, though: some fields have already woven AI into core workflows, while others still need stronger data, skills, and processes to capture its full value.

Despite notable advances in AI-enabled skincare, current systems remain limited in addressing individualized concerns. Most solutions target narrow components, such as image-based skin analysis or static ingredient matching, rather than a holistic, adaptive framework that continuously refines recommendations based on user feedback and contextual factors such as routine, products used together, seasonality, and constraints.

To our knowledge, no unified tool aggregates the skincare product landscape while treating user experience as primary evidence. By user experience we mean post-use review signals, stated concerns, routine context, and constraints—without which recommendations remain generic and hard to trust.

Large Language Models (LLMs) are state of the art for analyzing such unstructured text, and Retrieval-Augmented Generation (RAG) grounds their outputs in current product, ingredient, and clinical guidance data, reducing hallucinations and enabling source-linked explanations. Together, LLMs and RAG support a comprehensive, evidence-based system that learns from comparable cases and explains why a product fits.

Adopting this approach helps users receive recommendations that are more precise for their specific concerns and constraints, safer through proactive flagging of potential irritants or contraindications, and more transparent thanks to explanations and citations. It also improves decision speed (less trial-and-error), builds trust through auditable reasoning, adapts quickly as products and formulations change, and can reflect practical preferences such as budget, fragrance tolerance, and texture. For brands and retailers, better matches can translate into higher satisfaction and fewer returns.

In conclusion, centering user experience as evidence—paired with LLMs for language understanding and RAG for grounding—promises a step change in skincare recommendation: more personalized, trustworthy, and responsive to real-world needs. It also offers a replicable blueprint for evidence-grounded, domain-specific recommendation that can extend beyond skincare.

2. Aims of the research

The main aim of this research is to develop a personalized skincare recommendation framework that integrates Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG). Unlike traditional approaches, this system will not rely solely on static product descriptions but will also use user-generated reviews and detailed product information. By combining these sources, the framework will generate recommendations that are not only more accurate but also tailored to the unique needs of everyone. The ultimate goal is to make decision-making for skincare both easier and faster, reducing the confusion caused by marketing-driven claims and generalized product suggestions.

Another important aim of this study is to evaluate the role of vector databases in improving the efficiency of the recommendation process. Vector databases enable faster and more precise retrieval of skincare-related knowledge, ensuring that recommendations are based on the most relevant and up-to-date information. Their use will be assessed in terms of performance and scalability, particularly in handling large amounts of user reviews and product data.

In addition, this research seeks to explore how the integration of RAG with LLMs introduces new capabilities for domain-specific personalization. While AI-powered recommendation systems already exist in industries such as e-commerce and healthcare, their application in skincare has remained fragmented. By adapting these technologies to skincare, the framework will bridge this gap and demonstrate how advanced AI can refine recommendations continuously through contextual understanding and user feedback.

Finally, the study will investigate how users perceive the usefulness, clarity, and trustworthiness of the system. Since transparency is a major concern in skincare, recommendations generated through RAG will be grounded in real product data and verified user experiences. By offering explainable and evidence-based suggestions, the framework aims to build stronger trust with users, making them more confident in their skincare choices.

By combining RAG, LLMs, and vector databases, the system will deliver transparent, accurate, and highly personalized product recommendations. Beyond improving efficiency and relevance, it will also enhance user trust by grounding suggestions in real experiences and data. Ultimately, this work seeks to contribute both academically and practically, offering a foundation for future AI-driven personalization in skincare and beyond.

3. Hypotheses and research questions

To guide this research, the following research questions and hypotheses have been formulated, focusing on the integration of Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs) for developing personalized skincare recommendations. These will serve as the foundation for examining how user-generated reviews, product information, and vector databases can be leveraged to enhance accuracy, efficiency, and trust in AI-driven recommendation systems.

Research questions:

1. How does the integration of LLMs with retrieval mechanisms improve the personalization of skincare product suggestions compared to traditional recommendation methods?
2. How does incorporating real user feedback through reviews affect the accuracy and trustworthiness of skincare recommendations generated by the system?
3. What role do vector databases play in improving the speed and accuracy of information retrieval in a RAG-based recommendation system?
4. To what extent can RAG-based systems improve the efficiency, diversity, and reliability of skincare recommendations?

Hypothesis:

1. A skincare recommendation system that integrates RAG with LLMs will reduce irrelevant or conflicting product suggestions compared to LLM-only systems.
2. Incorporating user-generated reviews into the RAG framework will improve the accuracy and trustworthiness of skincare recommendations.
3. The integration of vector databases into a RAG-based skincare system will significantly improve retrieval speed and information accuracy compared to non-vector retrieval approaches.
4. RAG-enhanced recommendation systems will reduce the mismatch between user skin concerns and product suggestions compared to traditional approaches.

4. Research Methodology

The study begins with a structured literature review covering recommender systems that use LLMs and/or RAG, with attention to problem framing, datasets, modeling choices, evaluation protocols, and common failure modes (hallucination, sparsity, cold start, safety). This review informs design choices, baselines, and metrics.

Data collection will be conducted through web scraping. Publicly available product and review data will be scraped from Ulta and Sephora to build a comprehensive corpus of the current skincare market. The dataset will include product identifiers, brand, category, price, size, INCI ingredient lists, marketing claims, usage instructions, and availability, as well as user reviews with ratings, timestamps, free-text narratives, and any self-reported attributes (e.g., skin type or concerns).

Post-collection, data will be cleaned and normalized (deduplication across retailers, brand/product name standardization, INCI parsing), and reviews will be processed for entities and aspects. After cleaning, the corpus will be embedded and indexed in a vector database with rich metadata to enable fast, filtered semantic retrieval for the RAG + LLM recommender.

The system will undergo a comprehensive evaluation of recommendation fit to stated skin type, concerns, and constraints, alongside category correctness and ingredient suitability with clear, defensible rationales. Multiple test scenarios will be executed across diverse profiles (e.g., acne-prone, melasma, sensitive skin) and request types (e.g., moisturizer, sunscreen) to assess consistency and coverage. Results will be spot-reviewed by a qualified dermatologist or licensed esthetician to validate clinical plausibility and safety.

This methodology enables a rigorous, practice-oriented evaluation of a RAG-enabled LLM skincare recommender—assessing accuracy, category correctness, ingredient safety, and explanation quality across diverse skin profiles.

5. Importance of the thesis

This thesis addresses a persistent gap in AI-powered skincare: most systems offer generic suggestions without clear evidence or explanations. By positioning user experience as primary evidence and unifying product, ingredient, and review data, it advances recommendations from opinion-like outputs to accountable guidance that can be examined, challenged, and improved.

The core contribution is the integration of retrieval-augmented generation with large language models to produce source-cited, evidence-based recommendations. This combination enables nuanced analysis of unstructured text (reviews, product descriptions, ingredient lists) while grounding conclusions in verifiable references, reducing hallucinations and increasing transparency.

For individual users, the system aims to deliver recommendations that truly fit their skin type, concerns, and constraints, with clear rationales that explain why each product is suitable. Safer choices are encouraged through proactive identification of potential irritants or contraindications, while concise explanations and citations build trust and shorten decision time.

For companies and retailers, more accurate matches can translate into higher customer satisfaction, stronger loyalty, and fewer returns. Aggregated insights from de-identified user feedback and review patterns can inform product development, inventory planning, and responsible marketing, all while maintaining auditable reasoning that supports compliance and brand credibility.

Beyond skincare, the thesis offers a reusable blueprint for evidence-grounded recommendation in domains where products evolve quickly and user reviews carry critical signals. By emphasizing transparency, safety, and accountability, it contributes to the broader AI field's shift from opaque predictions to trustworthy, explainable systems that serve both consumers and businesses.

6. Structure of the thesis

Chapter 1: Abstract & Introduction

Summarizes the problem of generic, low-trust skincare recommendations and introduces a RAG-enabled LLM approach grounded in product, ingredient, and user-review evidence. States objectives, motivation, and core contributions.

Chapter 2: Literature Review

Synthesizes prior work on skincare and health recommenders, LLMs, RAG, and user-generated reviews to highlight gaps in personalization, safety, and explainability. Establishes why an evidence-grounded, case-based approach is needed.

Chapter 3: Methodology

Outlines data collection via web scraping, cleaning and normalization, vector indexing, and a retrieval-then-generation workflow with safety checks and transparent rationales. Describes a high-level evaluation plan focused on fit, category correctness, safety, and explanation quality.

Chapter 4: Results

Presents dataset statistics and key evaluation outcomes on recommendation accuracy, category correctness, ingredient safety, and explanation clarity. Includes illustrative examples of successes and edge cases.

Chapter 5: Findings and Discussion

Interprets results, explaining where evidence grounding improved specificity and trust and where limitations remain (e.g., sparse reviews or shifting formulations). Discusses implications for users, brands, and future deployments.

Chapter 6: Conclusion

Recaps contributions and practical impact: more personalized, transparent, and safer skincare recommendations. Notes limitations and outlines future work to strengthen robustness and broaden coverage.

7. Conclusion

This thesis demonstrates that centering real user experience as evidence—rather than relying on coarse profiles—can materially improve the quality and trustworthiness of skincare recommendations. By framing the problem around personalization, safety, and transparency, it addresses a clear gap in existing tools.

Conceptually, the work shows how language understanding coupled with grounded retrieval can turn noisy, unstructured reviews and product information into accountable guidance with clear, source-linked rationales. This shifts recommendations from opaque suggestions to explanations that can be audited and refined.

Practically, the approach benefits users through better fit to concerns and constraints and benefits companies through higher satisfaction and fewer returns. The emphasis on explainability helps build confidence, supports responsible marketing, and creates a foundation for compliance and continuous improvement.

Looking ahead, the framework invites broader validation with clinical experts, expansion to additional retailers and languages, and ongoing checks for bias and fairness. These steps can extend the method’s reach beyond skincare, offering a reusable blueprint for evidence-grounded recommendation in other consumer domains.

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